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Notched head design for tape applications

Abstract

In a tape drive comprising a tape head module, a tape contacts the tape head module during operation. The tape head module comprises a substrate, a plurality of data heads disposed adjacent to the substrate at a media facing surface (MFS), and a closure disposed adjacent to the plurality of data heads. The closure comprises a first side portion disposed at a leading edge, a second side portion disposed at the leading edge, the first and second side portions being recessed from the MFS, and a central portion disposed between the first and second side portions at the leading edge. In some embodiments, the central portion is recessed from the MFS. The central portion may comprise a notch, a taper, or one or more steps. The first and second side portions may be tapered, rounded, or comprise one or more steps to reduce contact of a tape during operation.

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Background/Summary

BACKGROUND OF THE DISCLOSURE

Field of the Disclosure

(1) Embodiments of the present disclosure generally relate to a tape drive comprising a tape head module.

Description of the Related Art

(2) Tape heads of tape drive systems are used to record and read back information on tapes by magnetic processes. Magnetic transducers of the tape heads read data from and write data onto magnetic recording media. Data is written on the magnetic recording media by moving a magnetic write transducer to a location over the media where the data is to be stored. The magnetic write transducer then generates a magnetic field, which encodes the data into the magnetic media. Data is read from the media by the magnetic read transducer through sensing of the magnetic field of the magnetic media.

(3) When writing data to or reading data from, the tape contacts the tape head or tape head module due to the roughness of the tape. However, such prolonged contact of the tape over the tape head module may cause friction or high contact force, damaging the tape. If the tape is damaged over time due to friction and/or the contact force exerted upon the tape, the data stored on the tape may be un-readable, and no new data may be written to the tape. As such, the overall lifetime of the tape is reduced.

(4) Therefore, there is a need in the art for a tape head module that reduces friction and contact force exerted upon a tape during operation.

SUMMARY OF THE DISCLOSURE

(5) In a tape drive comprising a tape head module, a tape contacts the tape head module during operation. The tape head module comprises a substrate, a plurality of data heads disposed adjacent to the substrate at a media facing surface (MFS), and a closure disposed adjacent to the plurality of data heads. The closure comprises a first side portion disposed at a leading edge, a second side portion disposed at the leading edge, the first and second side portions being recessed from the MFS, and a central portion disposed between the first and second side portions at the leading edge. In some embodiments, the central portion is recessed from the MFS. The central portion may comprise a notch, a taper, or one or more steps. The first and second side portions may be tapered, rounded, or comprise one or more steps to reduce contact of a tape during operation.

(6) In one embodiment, a tape head module comprises a substrate, a plurality of data heads disposed adjacent to the substrate at a media facing surface (MFS), the plurality of data heads being one or more of write heads and read heads, and a closure disposed adjacent to the plurality of data heads, the closure comprising: a first side portion disposed at a leading edge, the first side portion being recessed from the MFS, a second side portion disposed at the leading edge, the second side portion being recessed from the MFS, and a central portion disposed between the first and second side portions at the leading edge, the central portion comprising a notch.

(7) In another embodiment, a tape head module comprises a substrate, a plurality of data heads disposed adjacent to the substrate at a media facing surface (MFS), the plurality of data heads being one or more of write heads and read heads, and a closure disposed adjacent to the plurality of data

heads, the closure comprising: a first side portion disposed at a leading edge, a second side portion disposed at the leading edge, wherein the first and second side portions each individually comprises a first step recessed from the MFS and a second step recessed from the MFS, and a central portion disposed between the first and second side portions at the leading edge, the central portion being at least partially recessed from the MFS.

(8) In yet another embodiment, a tape head module comprises a substrate, a plurality of data heads disposed adjacent to the substrate at a media facing surface (MFS), the plurality of data heads being one or more of write heads and read heads, and a closure disposed adjacent to the plurality of data heads, the closure comprising: a first side portion disposed at a leading edge, a second side portion disposed at the leading edge, wherein the first and second side portions each individually comprises a first step recessed from the MFS and a second step recessed from the MFS, and a central portion disposed between the first and second side portions at the leading edge, the central portion comprising a third step recessed from the MFS and a fourth step recessed from the MFS.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of this disclosure and are therefore not to be considered limiting of its scope, for the disclosure may admit to other equally effective embodiments.

(2) FIGS. 1A-1C illustrate a perspective exploded view, a simplified top down, and side profile view of a tape drive, in accordance with some embodiments.

(3) FIG. 2 is a schematic illustration of a tape head and tape that are aligned.

(4) FIG. 3A illustrates a magnetic recording head according to one approach of the present disclosure that includes first, second and third modules each having a media facing surface, respectively, which may be flat, contoured, etc.

(5) FIG. 3B illustrates a magnetic recording head according to another approach of the present disclosure that includes first and second modules.

(6) FIGS. 4A-4C illustrate various configurations for the first, second and third modules of FIG. 3A.

(7) FIGS. 4D-4E illustrate various configurations for the first and second modules of FIG. 3B.

(8) FIG. 5A illustrates a media facing surface (MFS) view of the module, according to one embodiment.

(9) FIG. 5B illustrates a perspective view of the module of FIG. 5A.

(10) FIG. 6 illustrates a perspective view of a module, according to another embodiment.

(11) FIG. 7 illustrates a perspective view of a module, according to yet another embodiment.

(12) FIG. 8A illustrates a top-view of a module, according to one embodiment.

(13) FIG. 8B illustrates a perspective view of the module of FIG. 8A.

(14) FIG. 9A illustrates a top-view of a module, according to another embodiment.

(15) FIG. 9B illustrates a perspective view of the module of FIG. 9A.

(16) FIG. 10A illustrates a top-view of a module, according to yet another embodiment.

(17) FIG. 10B illustrates a perspective view of the module of FIG. 10A.

(18) FIG. 11A illustrates a top-view of a module, according to another embodiment.

(19) FIG. 11B illustrates a perspective view of the module of FIG. 11A.

(20) FIG. 12 illustrates a graphical representation of a tape moving over the module of FIGS. 8A-8B, according to one embodiment.

(21) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements disclosed in one embodiment may be beneficially utilized on other embodiments without specific recitation.

DETAILED DESCRIPTION

(22) In the following, reference is made to embodiments of the disclosure. However, it should be understood that the disclosure is not limited to specific described embodiments. Instead, any combination of the following features and elements, whether related to different embodiments or not, is contemplated to implement and practice the disclosure. Furthermore, although embodiments of the disclosure may achieve advantages over other possible solutions and/or over the prior art, whether or not a particular advantage is achieved by a given embodiment is not limiting of the disclosure. Thus, the following aspects, features, embodiments and advantages are merely illustrative and are not considered elements or limitations of the appended claims except where explicitly recited in a claim(s). Likewise, reference to “the disclosure” shall not be construed as a generalization of any inventive subject matter disclosed herein and shall not be considered to be an element or limitation of the appended claims except where explicitly recited in a claim(s).

(23) In a tape drive comprising a tape head module, a tape contacts the tape head module during operation. The tape head module comprises a substrate, a plurality of data heads disposed adjacent to the substrate at a media facing surface (MFS), and a closure disposed adjacent to the plurality of data heads. The closure comprises a first side portion disposed at a leading edge, a second side portion disposed at the leading edge, the first and second side portions being recessed from the MFS, and a central portion disposed between the first and second side portions at the leading edge. In some embodiments, the central portion is recessed from the MFS. The central portion may comprise a notch, a taper, or one or more steps. The first and second side portions may be tapered, rounded, or comprise one or more steps to reduce contact of a tape during operation.

(24) FIGS. 1A-1C illustrate a perspective exploded view, a simplified top down, and side profile view of a tape drive **100**, in accordance with some embodiments. The tape drive **100** may be a captive tape drive or a tape embedded drive (TED). Focusing on FIG. 1B, for example, the tape drive comprises a casing **105**, one or more tape reels **110**, one or more motors (e.g., a stepping motor **120** (also known as a stepper motor), a voice coil motor (VCM) **125**, etc.) a head assembly **130** with one or more read heads and one or more write heads, and tape guides/rollers **135a**, **135b**. In the descriptions herein, the term “head assembly” may be referred to as “magnetic recording head”, interchangeably, for exemplary purposes. Focusing on FIG. 1C, for example, the tape drive further comprises a printed circuit board assembly (PCBA) **155**. In an embodiment, most of the components are within an interior cavity of the casing, except the PCBA **155**, which is mounted on an external surface of the casing **105**. The same components are illustrated in a perspective view in FIG. 1A. In the descriptions herein, the term “tape” may be referred to as “magnetic media”, interchangeably, for exemplary purposes.

(25) In the illustrated embodiments, two tape reels **110** are placed in the interior cavity of the casing **105**, with the center of the two tape reels **110** on the same level in the cavity and with the head assembly **130** located in the middle and below the two tape reels **110**. Tape reel motors located in the spindles of the tape reels **110** can operate to wind and unwind the tape media **115** in the tape reels **110**. Each tape reel **110** may also incorporate a tape folder to help the tape media **115** be neatly wound onto the reel **110**. One or more of the tape reels **110** may form a part of a removable cartridge and are not necessarily part of the tape drive **100**. In such embodiments, the tape drive **100** may not be a tape embedded drive as it does not have embedded media, the drive **100** may instead be a tape drive configured to accept and access magnetic media or tape media **115** from an insertable cassette or cartridge (e.g., an LTO drive), where the insertable cassette or cartridge further comprises one or more of the tape reels **110** as well. In such embodiments, the tape or media **115** is contained in a cartridge that is removable from the drive **100**. The tape media **115** may be

made via a sputtering process to provide improved areal density. The tape media **115** comprises two surfaces, an oxide side and a substrate side. The oxide side is the surface that can be magnetically manipulated (written to or read from) by one or more read/write heads. The substrate side of the tape media **115** aids in the strength and flexibility of the tape media **115** by providing bending and extensional stiffness to the tape. The tape media **115** operates under a given tension in its longitudinal direction.

(26) Tape media **115** from the tape reels **110** are biased against the guides/rollers **135a**, **135b** (collectively referred to as guides/rollers **135**) and are movably passed along the head assembly **130** by movement of the reels **110**. The illustrated embodiment shows four guides/rollers **135a**, **135b**. The guide rollers **135a**, **135b** help maintain the tape aligned relative to head assembly **130** and reels **110**. The guide rollers **135b** are not co-linear with head assembly **130**. The shallow angle formed by the guide rollers **135a**, **135b** is called the wrap angle. The tape **115** is pressed against the head assembly **130** by virtue of the tape tension and the angle that results from the guide rollers **135b** and head assembly **130** not being co-linear.

(27) As shown in FIG. **1A**, in some embodiments, the guides/rollers **135** utilize the same structure. In other embodiments, as shown in FIG. **1B**, the guides/rollers **135** may have more specialized shapes and differ from each other based on function. Furthermore, a lesser or a greater number of rollers may be used. For example, the two function rollers may be cylindrical in shape, while the two functional guides may be flat-sided (e.g., rectangular prism) or clip shaped with two prongs and the film moving between the prongs of the clip.

(28) The voice coil motor **125** and stepping motor **120** may variably position the tape head(s) transversely with respect to the width of the recording tape. The stepping motor **120** may provide coarse movement, while the voice coil motor **125** may provide finer actuation of the head(s). In an embodiment, servo data may be written to the tape media to aid in more accurate position of the head(s) along the tape media **115**.

(29) In addition, the casing **105** comprises one or more particle filters **141** and/or desiccants **142**, as illustrated in FIG. **1A**, to help maintain the environment in the casing. For example, if the casing is not airtight, the particle filters may be placed where airflow is expected. The particle filters and/or desiccants may be placed in one or more of the corners or any other convenient place away from the moving internal components. For example, the moving reels may generate internal airflow as the tape media winds/unwinds, and the particle filters may be placed within that airflow.

(30) There is a wide variety of possible placements of the internal components of the tape drive **100** within the casing **105**. In particular, as the head mechanism is internal to the casing in certain embodiments, the tape media **115** may not be exposed to the outside of the casing **105**, such as in conventional tape drives. Thus, the tape media **115** does not need to be routed along the edge of the casing **105** and can be freely routed in more compact and/or otherwise more efficient ways within the casing **105**. Similarly, the head(s) **130** and tape reels **110** may be placed in a variety of locations to achieve a more efficient layout, as there are no design requirements to provide external access to these components.

(31) As illustrated in FIG. **1C**, the casing **105** comprises a cover **150** and a base **145**. The PCBA **155** is attached to the bottom, on an external surface of the casing **105**, opposite the cover **150**. As the PCBA **155** is made of solid state electronics, environmental issues are less of a concern, so it does not need to be placed inside the casing **105**. That leaves room inside casing **105** for other components, particularly, the moving components and the tape media **115** that would benefit from a more protected environment.

(32) In some embodiments, the tape drive **100** is sealed. Sealing can mean the drive is hermetically sealed or simply enclosed without necessarily being airtight. Sealing the drive may be beneficial for tape film winding stability, tape film reliability, and tape head reliability. Desiccant may be used to limit humidity inside the casing **105**.

(33) In one embodiment, the cover **150** is used to hermetically seal the tape drive. For example, the

drive **100** may be hermetically sealed for environmental control by attaching (e.g., laser welding, adhesive, etc.) the cover **150** to the base **145**. The drive **100** may be filled by helium, nitrogen, hydrogen, or any other typically inert gas.

(34) In some embodiments, other components may be added to the tape drive **100**. For example, a pre-amp for the heads may be added to the tape drive. The pre-amp may be located on the PCBA **155**, in the head assembly **130**, or in another location. In general, placing the pre-amp closer to the heads may have a greater effect on the read and write signals in terms of signal-to-noise ratio (SNR). In other embodiments, some of the components may be removed. For example, the filters **141** and/or the desiccant **142** may be left out.

(35) In various embodiments, the drive **100** includes controller **140** integrated circuits (IC) (or more simply “a controller **140**”) (e.g., in the form of one or more System on Chip (SoC)), along with other digital and/or analog control circuitry to control the operations of the drive. For example, the controller **140** and other associated control circuitry may control the writing and reading of data to and from the magnetic media, including processing of read/write data signals and any servo-mechanical control of the media and head module. In the description below, various examples related to writing and reading and verifying of written data, as well as control of the tape head and media to achieve the same, may be controlled by the controller **140**. As an example, the controller **140** may be configured to execute firmware instructions for the various same gap verify embodiments described below.

(36) FIG. 2 is a schematic illustration of a tape head module **200** and a tape **204** that are aligned. The tape head module **200** comprises a tape head body **202** that is aligned with the tape **204**. The tape **204** moves past the tape head module **200** during read and/or write operations. The tape head module **200** has a media facing surface (MFS) **214** that faces the tape **204**. The tape head module **200** is coupled to a controller, which may be the controller **140** of FIG. 1.

(37) The tape head body **202** comprises a first servo head **206A** and a second servo head **206B** spaced therefrom. It is to be understood that while two servo heads have been shown, the disclosure is not limited to two servo heads. Rather, it is contemplated that more or less servo heads may be present. A plurality of data heads **208A-208G** is disposed between the first servo head **206A** and the second servo head **206B**. It is to be understood that while seven data heads have been shown, the disclosure is not limited to seven data heads. Rather, the number of data heads can be more or less than seven, depending on the requirements of the embodiment. For example there can be sixteen, thirty two, sixty four or more data heads utilized in the tape head body **202**.

(38) A plurality of pads **220A-220N** is electrically coupled to the data head body **202**. The plurality of pads **220A-220N** coupled to the data head body **202** is not limited to the number shown in FIG. 2. Rather, more or less pads are contemplated. The pads **220A-220N** are used to connect the drive electronics to the servo heads **206A**, **206B** and to data read and writer elements. The pads **220A-220N** are used to establish the potential across the servo reader by means of a power supply (not shown) embedded in the tape head **200**.

(39) The tape **204** comprises a first servo track **210A** and a second servo track **210B**. The first servo track **210A** and the second servo track **210B** are spaced apart allowing the tape head **200** to monitor and control the average position of the data heads **208A-208G** relative to the data tracks **212A-212G** on the tape **204**. It is to be understood that while two servo tracks have been shown, the disclosure is not limited to two servo tracks. Rather, the number of servo tracks can be more or less than two, depending on the requirements of the embodiment.

(40) The tape **204** further comprises a plurality of data tracks **212A-212G** disposed between the first servo track **210A** and the second servo track **210B**. It is to be understood that while seven data tracks have been shown, the disclosure is not limited to the seven data tracks. Rather, the number of data tracks can be more or less than seven, depending on the requirements of the embodiment. In the embodiment of FIG. 2, the first servo head **206A** reads its lateral position information (e.g., alignment) over the first servo track **210A**. The second servo head **206B** is aligned with the second

servo track **210B**. The combined information allows the servo actuator of the tape drive **200** to align the data heads **208A-208G** such that the center data head (e.g., **208D**) is centered on tape **204**. The plurality of data heads **208A-208G** is thus individually aligned with the plurality of data tracks **212A-212N** for best case positioning. In this embodiment the first servo head **206A**, the second servo head **206B**, the first servo track **210A**, the second servo track **210B**, the plurality of data heads **208A-208G**, and the plurality of data tracks **212A-212G** are able to read and/or write the data accurately because all are aligned perpendicular to the direction of travel of the tape **204**.

(41) FIG. 3A illustrates a magnetic recording head **126** according to one approach of the present disclosure that includes first, second and third modules **302**, **304**, **306** each having a media facing surface **308**, **310**, **312** respectively, which may be flat, contoured, etc. FIGS. 4A-4C illustrate various configurations for the first, second and third modules **302**, **304**, **306** of FIG. 3A.

(42) For explanation purposes, the first module **302**, or the leading module **302**, is the first module encountered by the tape **315** in a three module design for tape **315** moving in the indicated direction by arrow **350**. The third module **306**, or the trailing module **306**, follows the middle module **304** and is the last module seen by the tape **315** in a three module design. The leading and trailing modules **302**, **306** are referred to collectively as outer modules. Also note that the outer modules **302**, **306** will alternate as leading modules, depending on the direction of travel of the tape **315**.

(43) In a write-read-write (W-R-W) head, outer modules for writing flank one or more inner modules for reading. FIG. 4A illustrates a W-R-W configuration, the outer modules **302**, **306** each include one or more arrays of writers (W) **260**. The inner module **304** of FIG. 4A includes one or more arrays of readers (R) **258** in a similar configuration. Variations of a multi-module head include a R-W-R head, like shown in FIG. 4B, a R-R-W head, a W-W-R head, etc. In yet other variations, one or more of the modules may have read/write pairs of transducers. While each module comprises 8 readers **258** or 8 writers **260**, each module may include a greater number of readers **258** and/or writers **260**. In other embodiments, like shown in FIG. 4C, each module **302**, **304**, **306** may individually be tilted at an angle of about ± 3 degrees to about ± 15 degrees, and each module **302**, **304**, **306** may be configured to dynamically tilt an angle of about ± 0.1 degrees to about ± 1 degree to adjust a track pitch to accommodate changes of a tape. The track pitch of the tape may vary due to creep, such as by temperature and storage duration. Tilting the modules **302**, **304**, **306** enables the modules **302**, **304**, **306** to accommodate for changes in the track pitch.

(44) Writing and reading functions are performed by different modules at any given time. In one approach, the second module **304** includes a plurality of readers (such as the readers **258** of FIGS. 4A-4C) and optional servo readers **329** and no writers. The first and third modules **302**, **306** include a plurality of writers **322** and no data readers, with the exception that the outer modules **302**, **306** may include optional servo readers. The writers **322** may be the writers **260** of FIGS. 4A-4C. The servo readers may be used to position the head during reading and/or writing operations. The servo reader(s) on each module are typically located towards the end of the array of readers or writers.

(45) In one approach, the media facing surfaces **308**, **310**, **312** of the first, second, and third modules **302**, **304**, **306** lie on about parallel planes, and the media facing surface **310** of the second module **304** is above the media facing surfaces **308**, **312** of the first and third modules **302**, **306**. Such a configuration has the effect of creating the desired wrap angle of the tape **315** relative to the media facing surface **310** of the second module **304**.

(46) Where the media facing surfaces **308**, **310**, **312** lie along parallel or nearly parallel yet offset planes, the volume contained between the edge **318** and the tape **315** increases downstream to balance the tape bending stiffness and tension. Such volume increment causes the pressure of the fluid contained therein to reduce below atmospheric level. This low pressure brings the tape **315** adhered close to the media facing surface **308** of the leading module **302**. A trailing edge **320** of the leading module **302** (the end from which the tape **315** leaves the leading module **302**) is the approximate reference point which defines a wrap angle over the media facing surface **310** on the

leading edge **311** of the second module **304**. The tape **315** stays in close proximity to the media facing surface **308** until close to the trailing edge **320** of the leading module **302**. Accordingly, the writers **322** may be located near the trailing edges of the outer modules **302**, **306**.

(47) Beneficially, the inner wrap angle on the leading edge **311** of the module **304** receiving the tape **315** will be larger than the inner wrap angle on the trailing edge **313**, as the tape **315** rides above the trailing module **306**. This difference is generally beneficial as a smaller inner wrap angle on the trailing edge **313** tends to oppose what has heretofore been a steeper exiting effective wrap angle.

(48) The media facing surfaces **308**, **312** of the outer modules **302**, **306** are positioned to achieve a negative wrap angle at the trailing edge **320** of the leading module **302**. This is generally beneficial in helping to reduce friction due to low contact pressure with the trailing end **320**. Further, at the trailing module **306**, the tape **315** flies over the media facing surface **312** with a relatively large clearance so there is virtually no contact pressure nor wear on the elements when tape is moving in this direction. Particularly, the tape **315** entrains air and so will not significantly ride on the media facing surface **312** of the third module **306** (some contact may occur). This is permissible, because the leading module **302** is writing while the trailing module **306** is idle.

(49) As shown in FIG. 3A, the first, second, and third modules **302**, **304**, **306** each have a closure **332**, **334**, **336**, and a substrate **331**, **333**, **335**, each of which extends the media facing surface of the associated module, thereby effectively positioning the read/write elements away from the edge of the media facing surface. The closures **332**, **334**, **336** can be a ceramic closure of a type typically found on tape head modules. The closures **334**, **336** of the first and third modules **302**, **306**, however, may be shorter than the closure **332** of the second module **304** as measured parallel to a direction of tape travel over the respective module. This enables positioning the modules closer together.

(50) With reduced-thickness ceramic or thin film closures **334**, **336** or no closures on the outer modules **302**, **306**, the write-to-read gap spacing can be reduced to less than about 1 mm, e.g., about 0.75 mm, or 50% less than commonly-used linear tape open (LTO) tape head spacing. The open space between the modules **302**, **304**, **306** can still be set to approximately 0.5 to 0.6 mm, which in some approaches is ideal for stabilizing tape motion over the second module **304**.

(51) FIG. 3B illustrates a magnetic recording head **300** according to another approach of the present disclosure that includes first and second modules **352**, **354**, each having a media facing surface **356**, **358**, respectively, which may be flat, contoured, etc. FIGS. 4D-4E illustrate various configurations for the first and second modules **352**, **354**, of FIG. 3B.

(52) The magnetic recording head **300** of FIG. 3B is similar to the magnetic recording head **126** of FIG. 3A; however, the magnetic recording head **300** comprises only a first module **352** and a second module **354**. The first module **352** comprises a closure **364** and a substrate **360**, where the top or upper surfaces of the closure **364** and the substrate **360** form the MFS **356**. The second module **354** comprises a closure **366** and a substrate **362**, where the top or upper surfaces of the closure **362** and the substrate **366** form the MFS **358**.

(53) The magnetic recording head **300** of FIG. 3B is similar to the magnetic recording head **126** of FIG. 3A; however, the magnetic recording head **300** comprises only a first module **352** and a second module **354**. The first module **352** comprises a closure **360** and a substrate **364**, where the top or upper surfaces of the closure **360** and the substrate **364** form the MFS **356**. The second module **354** comprises a closure **362** and a substrate **366**, where the top or upper surfaces of the closure **362** and the substrate **366** form the MFS **358**.

(54) As shown in the MFS views of FIGS. 4D and 4E, the modules **352**, **354** each individually comprise a plurality of data heads **368** or data elements. Each data head **368** comprises a write head and a read head. Thus, each module **352**, **354** is able to both write data to and read data from a tape. While each module comprises 4 data heads **368**, each module may include a greater number of readers **258** and/or writers **260**.

(55) In some embodiments, when a tape moves over the magnetic recording head **300** in a first direction, the first module **352** is controllable to write data to the tape while the second module **354** is controllable to read data from the tape. When the tape moves over the magnetic recording head **300** in a second direction opposite the first direction, the second module **354** is controllable to write data to the tape while the first module **352** is controllable to read data from the tape. Each module **352**, **354** may be tilted at an angle of about ± 3 degrees to about ± 15 degrees, as shown in FIG. 4E, and each module **352**, **354** may be configured to dynamically tilt an angle of about ± 0.1 degrees to about ± 1 degree to adjust a track pitch to accommodate changes of a tape. The track pitch of the tape may vary due to creep, such as by temperature and storage duration. Tilting the modules **352**, **354** enables the modules **352**, **354** to accommodate for changes in the track pitch.

(56) FIGS. 5A-11B illustrate modules **500**, **600**, **700**, **800**, **900**, **1000**, **1100**, of tape head assemblies, according to various embodiments. Each module **500**, **600**, **700**, **800**, **900**, **1000**, **1100** may be a module **302**, **304**, and/or **306** of the magnetic recording head **126** of FIG. 3A. Each module **500**, **600**, **700**, **800**, **900**, **1000**, **1100** may be a module **352** and/or **354** of the magnetic recording head **300** of FIG. 3B. Aspects of the modules may be used with one another.

(57) Each module **500**, **600**, **700**, **800**, **900**, **1000**, **1100** comprises a closure **502**, **602**, **702**, **802**, **902**, **1002**, **1102**, respectively, a substrate **504**, **604**, **704**, **804**, **904**, **1004**, **1104**, respectively, and a recession **506**, **606**, **706**, **806**, **906**, **1006**, **1106**, respectively. The read heads and/or write heads (not shown) of each module **500**, **600**, **700**, **800**, **900**, **1000**, **1100** are disposed within the recession **506**, **606**, **706**, **806**, **906**, **1006**, **1106**.

(58) The closures **502**, **602**, **702**, **802**, **902**, **1002**, **1102** may be a closure **332**, **334**, **336** of the magnetic recording head **126** of FIG. 3A. The substrates **504**, **604**, **704**, **804**, **904**, **1004**, **1104** may be a substrates **331**, **333**, and/or **335** of the magnetic recording head **126** of FIG. 3A. The closures **502**, **602**, **702**, **802**, **902**, **1002**, **1102** may be a closure **364**, **366** of the magnetic recording head **300** of FIG. 3B. The substrates **504**, **604**, **704**, **804**, **904**, **1004**, **1104** may be a substrates **364**, **366**, of the magnetic recording head **300** of FIG. 3B. In some embodiments, aspects of the closures **502**, **602**, **702**, **802**, **902**, **1002**, **1102** may be apply to the substrates **504**, **604**, **704**, **804**, **904**, **1004**, **1104**. While not explicitly described, a substrate can have the shaping features in the disclosed closures described below. In embodiments comprising one or more modules, such as described in FIGS. 3A-3B and 4A-4E, each closure and/or substrate may have the features of the closures **502**, **602**, **702**, **802**, **902**, **1002**, **1102** described below.

(59) FIG. 5A illustrates a media facing surface (MFS) view of the module **500**, according to one embodiment. FIG. 5B illustrates a perspective view of the module **500** of FIG. 5A, according to one embodiment.

(60) The closure **502** comprises a top surface **502a** disposed at the MFS, a first side region **503a**, a second side region **503b**, and a central region **501** disposed between the first and second side regions **503a**, **503b**. The central region **501** and the side regions **503a**, **503b** may be disposed at a leading edge of the closure **502** (and/or substrate **504**) and/or a trailing edge of the closure **502** (and/or substrate **504**). The top surface **502a** is substantially flat or planar. The closure **502** has a depth **510** in the z-direction of about 10 μm to about 150 μm . Each side region **503a**, **503b** has a length **514** in the x-direction of about 4 mm to about 10 mm. The central region **501** has a length **512** in the x-direction of about 0.8 mm to about 5 mm, such as about 3 mm. The closure **502** has a total length in the x-direction of about 20 mm to about 25 mm.

(61) The central region **501** is the active region where the read heads and/or write heads of the module **500** are located. The central region **501** comprises a cutout, via, or notch **508**, as shown in FIG. 5B, where the notch **508** has the same length **512** as the central region **501**. The notch **508** is disposed at an angle of about 5 degrees to about 90 degrees with respect to the MFS. As further shown in FIG. 5B, the side regions **503a**, **503b** (collectively referred to herein as side regions **503**) are both tapered or beveled such that the side regions **503** are angled downwards with respect to the MFS at an angle of about 0.1 degrees to about 5 degrees with respect to the MFS.

(62) During operation, a tape (not shown) moves over the module in the z-direction and/or the -z-direction. As the tape moves over the module **500**, the tapered side regions **503** increase the clearance between the side regions **503** and the tape while the notch **508** reduces the clearance between the tape and the central region **501** (i.e., where the read heads and/or write heads are located in the recession **506**). The notch **508** of the central region **501** does not pressurize as much as the side regions **503**, thus reducing the fly height or clearance of the tape over the central region **501**. For example, the clearance between the tape and the side regions **503** in the y-direction is greater than about 100 nm, whereas the clearance between the tape and the central region **501** in the y-direction is about 10 nm. By utilizing the tapered side regions **503** and the notch **508** of the central region **501**, friction and contact force exerted on the tape is minimized while the tape-to-head MFS clearance in the active region reaches an optimum level.

(63) While only the closure **502** is shown as comprising the tapered side regions **503** and notch **508** of the central region **501**, in some embodiments, the substrate **504** may instead comprise the tapered side regions **503** and notch **508**. In other embodiments, both the closure **502** and the substrate **504** may comprise tapered side regions **503** and the notch **508**.

(64) FIG. **6** illustrates a perspective view of a module **600**, according to another embodiment. The module **600** is similar to the module **500** of FIGS. **5A-5B**; however, the side regions **603a**, **603b** of the module **600** are rounded, rather than tapered. The central region **601** comprises the notch **508**, and the closure **602** comprises the substantially flat or planar top surface **602a** disposed at the MFS. The side regions **603a**, **603b** have a radius of curvature of about 0.1 mm to about 10 mm. The central region **601** and the side regions **603a**, **603b** may be disposed at a leading edge of the closure **602** (and/or substrate **604**) and/or a trailing edge of the closure **602** (and/or substrate **604**).

(65) As the tape moves over the module **600**, the curved side regions **603a**, **603b** allow more air flow to enter under the tape and generate more pressure that increases the clearance between the side regions **603a**, **603b** and the tape while the notch **508** reduces the clearance between the tape and the central region **601** (i.e., where the read heads and/or write heads are located in the recession **606**). The notch **508** of the central region **601** reduces the pressurization of the incoming air, thus reducing the fly height or clearance of the tape over the central region **601**. For example, the clearance between the tape and the side regions **603a**, **603b** in the y-direction is greater than about 100 nm, whereas the clearance between the tape and the central region **601** in the y-direction is about 10 nm. By utilizing the curved side regions **603a**, **603b** and the notch **508** of the central region **601**, friction and contact force exerted on the tape is minimized.

(66) FIG. **7** illustrates a perspective view of a module **700**, according to yet another embodiment. The module **700** comprises a closure **702**, which comprises a top surface **702a** disposed at the MFS, a first side region **703a**, a second side region **703b**, and a central region **701** disposed therebetween. The central region **701** comprises a cutout or notch **708**. The notch **708** is recessed by deep etching a distance **728** of about 0.5 μm to about 5 μm from the MFS. Each side region **703a**, **703b** comprises a first etched step **720a** recessed a distance **724** of about 0.3 μm to about 1.0 μm from the MFS in the y-direction and a second etched step **720b** recessed a distance **726** of about 0.5 μm to about 2.5 μm from the MFS in the y-direction. In other words, the side regions **703a**, **703b** comprise etched step-like features. Each step **720a**, **720b** has a width **725** in the z-direction of about 20 μm to about 150 μm . The central region **701** and the side regions **703a**, **703b** may be disposed at a leading edge of the closure **702** (and/or substrate **704**) and/or a trailing edge of the closure **702** (and/or substrate **704**).

(67) As the tape moves over the module **700**, the etched step-like side regions **703a**, **703b** increase the clearance between the side regions **703a**, **703b** and the tape while the etched notch **708** reduces the clearance between the tape and the central region **701** (i.e., where the read heads and/or write heads are located in the recession **706**). The notch **708** of the central region **701** reduces the pressurization of the incoming air, thus reducing the fly height or clearance of the tape over the central region **701**. For example, the clearance between the tape and the side regions **703a**, **703b** in

the y-direction is greater than about 100 nm, whereas the clearance between the tape and the central region **701** in the y-direction is about 10 nm. By utilizing the etched step-like side regions **703a**, **703b** and the etched notch **708** of the central region **701**, friction and contact force exerted on the tape is minimized.

(68) FIG. **8A** illustrates a top-view of a module **800**, according to one embodiment. FIG. **8B** illustrates a perspective view of the module **800** of FIG. **8A**. The module **800** is similar to the module **700** of FIG. **7**; however, the central region **801** is not recessed from the MFS. Rather, the central region **801** comprises a notch **808** disposed at the MFS and aligned with the top surface **802a** of the module **800**. The notch **808** is substantially flat or planar. Each side region **803a**, **803b** comprises the first step **720a** recessed the distance **724** of about 0.2 μm to about 2.5 μm from the MFS in the y-direction and the second step **720b** recessed the distance **726** of about 0.5 μm to about 5 μm from the MFS in the y-direction. In other words, the side regions **803a**, **803b** comprise etched step-like features. Each step **720a**, **720b** has a width **725** in the z-direction of about 10 μm to about 150 μm . The central region **801** and the side regions **803a**, **803b** may be disposed at a leading edge of the closure **802** (and/or substrate **804**) and/or a trailing edge of the closure **802** (and/or substrate **804**).

(69) As the tape moves over the module **800**, the stair-like side regions **803a**, **803b** pressure the air and increase the clearance between the side regions **803a**, **803b** and the tape while the notch **808** reduces the pressure and the clearance between the tape and the central region **801** (i.e., where the read heads and/or write heads are located in the recession **806**). The notch **808** of the central region **801** reduces the pressurization of the air, thus reducing the fly height or clearance of the tape over the central region **801**. For example, the clearance between the tape and the side regions **803a**, **803b** in the y-direction is greater than about 100 nm, whereas the clearance between the tape and the central region **801** in the y-direction is about 10 nm. By utilizing the stair-like side regions **803a**, **803b** and the notch **808** of the central region **801**, friction and contact force exerted on the tape is minimized.

(70) FIG. **9A** illustrates a top-view of a module **900**, according to another embodiment. FIG. **9B** illustrates a perspective view of the module **900** of FIG. **9A**. The module **900** is similar to the module **800** of FIGS. **8A-8B**; however, the central region **901** comprises a notch **908** disposed adjacent to and aligned with the top surface **902a** of the closure **902** and a tapered portion **918**. The tapered portion **918** extends from the notch **908** (i.e., the MFS) to a surface opposite the MFS at an angle of about 0.1 degrees to about 2 degrees, such as about 0.25 degrees, with respect to the MFS. The notch **908** is substantially flat or planar and has a width **930** in the z-direction of about 0 μm to about 100 μm (i.e., the notch **908** may be excluded in some embodiments).

(71) Each side region **903a**, **903b** comprises the first step **720a** recessed the distance **724** of about 0.2 μm to about 1.5 μm from the MFS in the y-direction and the second step **720b** recessed the distance **726** of about 0.5 μm to about 5 μm from the MFS in the y-direction. In other words, the side regions **903a**, **903b** comprise etched step-like features. Each step **720a**, **720b** has a width **725** in the z-direction of about 10 μm to about 100 μm . The central region **901** and the side regions **903a**, **903b** may be disposed at a leading edge of the closure **902** (and/or substrate **904**) and/or a trailing edge of the closure **902** (and/or substrate **904**).

(72) As the tape moves over the module **900**, the stair-like side regions **903a**, **903b** pressurize the air and increase the clearance between the side regions **903a**, **903b** and the tape while the notch **908** and tapered portion **918** reduce the clearance between the tape and the central region **901** (i.e., where the read heads and/or write heads are located in the recession **906**). The notch **908** and tapered portion **918** of the central region **901** reduces the pressurization of incoming air, thus reducing the fly height or clearance of the tape over the central region **901**. For example, the clearance between the tape and the side regions **903a**, **903b** in the y-direction is greater than about 100 nm, whereas the clearance between the tape and the central region **901** in the y-direction is about 10 nm. By utilizing the stair-like side regions **903a**, **903b** and the notch **908** and tapered

portion **918** of the central region **901**, friction and contact force exerted on the tape is minimized.

(73) FIG. **10A** illustrates a top-view of a module **1000**, according to yet another embodiment. FIG. **10B** illustrates a perspective view of the module **1000** of FIG. **10A**. The module **1000** is similar to the module **900** of FIGS. **9A-9B**; however, the central regions **1001** comprises the notch **908** disposed adjacent to and aligned with the top surface **1002a** of the closure **1002** and a third step **1020a** and a fourth step **1020b**. The notch **908** is substantially flat or planar and has a width **930** in the z-direction of about 0 μm to about 50 μm (e.g., the notch **908** may be excluded in some embodiments). The third step **1020a** is recessed a distance **1024** in the y-direction of about 0.05 μm to about 0.5 μm , such as about 0.1 μm , from the MFS. The fourth step **1020b** is recessed a distance **1026** in the y-direction of about 0.1 μm to about 1.0 μm , such as about 0.185 μm , from the MFS. Each step **1020a**, **1020b** has a width **1025** in the z-direction of about 0.1 μm to about 50 μm .

(74) Each side region **1003a**, **1003b** comprises the first step **720a** recessed the distance **724** of about 0.1 μm to about 2 μm from the MFS in the y-direction and the second step **720b** recessed the distance **726** of about 0.5 μm to about 5 μm from the MFS in the y-direction. Thus, the third and fourth steps **1020a**, **1020b** are recessed a smaller distance **1024**, **1026** from the MFS than the first and second steps **720a**, **720b** of the side regions **1003a**, **1003b**. Each step **720a**, **720b** has a width **725** in the z-direction of about 10 μm to about 100 μm . The central region **1001** and the side regions **1003a**, **1003b** may be disposed at a leading edge of the closure **1002** (and/or substrate **1004**) and/or a trailing edge of the closure **1002** (and/or substrate **1004**).

(75) As the tape moves over the module **1000**, the etched step-like side regions **1003a**, **1003b** pressurize the air and increase the clearance between the side regions **1003a**, **1003b** and the tape while the notch **908** and steps **1020a**, **1020b** reduce the clearance between the tape and the central region **1001** (i.e., where the read heads and/or write heads are located in the recession **1006**). The notch **908** and steps **1020a**, **1020b** of the central region **1001** reduces the pressurization of incoming air, thus reducing the fly height or clearance of the tape over the central region **1001**. For example, the clearance between the tape and the side regions **1003a**, **1003b** in the y-direction is greater than about 100 nm, whereas the clearance between the tape and the central region **1001** in the y-direction is about 10 nm. By utilizing the stair-like side regions **1003a**, **1003b** and the notch **908** and steps **1020a**, **1020b** of the central region **1001**, friction and contact force exerted on the tape is minimized.

(76) FIG. **11A** illustrates a top-view of a module **1100**, according to another embodiment. FIG. **11B** illustrates a perspective view of the module **1100** of FIG. **11A**. The module **1100** is similar to the module **1000** of FIGS. **10A-10B**; however, the central region **1101** does not comprise the notch **908** disposed adjacent to and aligned with the top surface **1102a** of the closure **1102**. Rather, the central region **1101** comprises a third step **1120a** and a fourth step **1120b**. The third step **1120a** is recessed a distance **1124** in the y-direction of about 0.5 μm to about 2.5 μm from the MFS. The fourth step **1120b** is recessed a distance **1126** in the y-direction of about 0.5 μm to about 2.5 μm from the MFS. Each step **1120a**, **1120b** has a width **1125** in the z-direction of about 10 μm to about 100 μm . Thus, comparing the third and fourth steps **1120a**, **1120b** of the closure **1102** to the third and fourth steps **1020a**, **1020b** of the closure **1002** of FIGS. **10A-10B**, the third and fourth steps **1120a**, **1120b** are recessed further from the MFS than the third and fourth steps **1020a**, **1020b**. Both third and fourth steps **1020a**, **1020b** can be made with a single mask and have common depths.

(77) Each side region **1103a**, **1103b** comprises the first step **720a** recessed the distance **724** of about 0.2 μm to about 1.5 μm from the MFS in the y-direction and the second step **720b** recessed the distance **726** of about 0.5 μm to about 5 μm from the MFS in the y-direction. Thus, the third and fourth steps **1120a**, **1120b** are recessed a greater distance **1124**, **1126** from the MFS than the first and second steps **720a**, **720b** of the side regions **1103a**, **1103b**. Each step **720a**, **720b** has a width **725** in the z-direction of about 10 μm to about 100 μm . The central region **1101** and the side regions **1103a**, **1103b** may be disposed at a leading edge of the closure **1102** (and/or substrate **1104**) and/or a trailing edge of the closure **1102** (and/or substrate **1104**).

(78) As the tape moves over the module **1100**, the step-like side regions **1103a**, **1103b** pressurizes the air and increase the clearance between the side regions **1103a**, **1103b** and the tape while steps **1120a**, **1120b** reduce the clearance between the tape and the central region **1101** (i.e., where the read heads and/or write heads are located in the recession **1106**). The steps **1120a**, **1120b** of the central region **1101** reduces the pressurization of incoming air, thus reducing the fly height or clearance of the tape over the central region **1101**. For example, the clearance between the tape and the side regions **1103a**, **1103b** in the y-direction is greater than about 100 nm, whereas the clearance between the tape and the central region **1101** in the y-direction is about 10 nm. By utilizing the step-like side regions **1103a**, **1103b** and the steps **1120a**, **1120b** of the central region **1101**, friction and contact force exerted on the tape is minimized.

(79) FIG. **12** illustrates a graphical representation **1200** of a tape **1202** moving over the module **800** of FIGS. **8A-8B** during operation, according to one embodiment. Only the first side region **803a** and the central region **801** of the module **800** are shown for clarity. Moreover, while only the module **800** of FIGS. **8A-8B**, any other modules **500**, **600**, **700**, **900**, **1000**, and **1100** would look similar. In the graphical representation **1200**, the y-axis represents the transversal direction (TD) of the tape **1202**, the x-axis represents the machine direction (MD) of the tape over the module **800**, and the z-axis represents an arbitrary unit (AU) of length in the z-direction.

(80) As shown in FIG. **12**, as the tape moves over the central region **801**, a first portion **1202a** of the tape **1202** is disposed closer to the MFS of the module **800** than a second portion **1202b** of the tape **1202** is disposed from the first side region **803a**. The close distance or low clearance of the first portion **1202a** of the tape **1202** over the central region **810** is needed for best magnetic recording. The further distance or high clearance of second portion **1202b** of the tape **1202** over the side regions **803a**, **803b** increases the reliability of the tape **1202** by reducing both the contact pressure and friction. The tape **1202** being disposed closer to the central region **801** than to the side region **803a** is due to the side region **803a** comprising the steps **720a**, **702b**, as the steps **720a**, **720b** reduce the pressurization of incoming air, thus reducing the fly height or clearance of the tape **1202** over the central region **801**.

(81) Therefore, by utilizing a closure and/or substrate having recessed side portion disposed on a leading edge, the fly height clearance of a tape is increased in the recessed side portions and decreased in a central portion of the leading edge of the closure. As such, friction and contact force exerted on the tape is minimized, ensuring the long-term reliability of the magnetic reader and writer.

(82) In one embodiment, a tape head module comprises a substrate, a plurality of data heads disposed adjacent to the substrate at a media facing surface (MFS), the plurality of data heads being one or more of write heads and read heads, and a closure disposed adjacent to the plurality of data heads, the closure comprising: a first side portion disposed at a leading edge, the first side portion being recessed from the MFS, a second side portion disposed at the leading edge, the second side portion being recessed from the MFS, and a central portion disposed between the first and second side portions at the leading edge, the central portion comprising a notch.

(83) The notch is recessed from the MFS. The notch is disposed at the MFS. The first and second side portions are tapered. The first and second side portions are disposed at an angle of about 0.1 degrees to about 5 degrees with respect to the MFS. The first and second side portions are rounded. The first and second side portions have a radius of curvature of about 0.1 mm to about 10 mm. A tape drive comprises the tape head module.

(84) In another embodiment, a tape head module comprises a substrate, a plurality of data heads disposed adjacent to the substrate at a media facing surface (MFS), the plurality of data heads being one or more of write heads and read heads, and a closure disposed adjacent to the plurality of data heads, the closure comprising: a first side portion disposed at a leading edge, a second side portion disposed at the leading edge, wherein the first and second side portions each individually comprises a first step recessed from the MFS and a second step recessed from the MFS, and a central portion

disposed between the first and second side portions at the leading edge, the central portion being at least partially recessed from the MFS.

(85) The central portion is entirely recessed from the MFS a distance of about 0.1 μm to about 10 μm . The central portion comprises a notch disposed at the MFS and a tapered portion recessed from the MFS. The tapered portion is disposed at an angle of about 0.1 degrees to about 0.5 degrees with respect to the MFS, and wherein the notch has a width of about 1 mm to about 6 mm. The first step is recessed a distance of about 0.2 μm to about 1 μm from the MFS, and wherein the second step is recessed a distance of about 0.5 μm to about 2.5 μm from the MFS. A tape drive comprises the tape head module.

(86) In yet another embodiment, a tape head module comprises a substrate, a plurality of data heads disposed adjacent to the substrate at a media facing surface (MFS), the plurality of data heads being one or more of write heads and read heads, and a closure disposed adjacent to the plurality of data heads, the closure comprising: a first side portion disposed at a leading edge, a second side portion disposed at the leading edge, wherein the first and second side portions each individually comprises a first step recessed from the MFS and a second step recessed from the MFS, and a central portion disposed between the first and second side portions at the leading edge, the central portion comprising a third step recessed from the MFS and a fourth step recessed from the MFS.

(87) The first and second steps are recessed further from the MFS than the third and fourth steps. The third and fourth steps are recessed further from the MFS than the first and second steps. The first step is recessed a distance of about 0.2 μm to about 1 μm from the MFS, and wherein the second step is recessed a distance of about 0.5 μm to about 2 μm from the MFS. The third step is recessed a distance of about 0.6 μm to about 2.5 μm from the MFS, and wherein the fourth step is recessed a distance of about 1 μm to about 13 μm from the MFS. The third step is recessed a distance of about 0.05 μm to about 0.15 μm from the MFS, and wherein the fourth step is recessed a distance of about 0.155 μm to about 0.2 μm from the MFS. A tape drive comprises the tape head module.

(88) While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A tape head module, comprising: a substrate; a plurality of data heads disposed adjacent to the substrate at a media facing surface (MFS), the plurality of data heads being one or more of write heads and read heads; and a closure disposed adjacent to the plurality of data heads, the closure comprising: a first side portion disposed at a leading edge, the first side portion being recessed from the MFS; a second side portion disposed at the leading edge, the second side portion being recessed from the MFS, wherein the first and second side portions each individually comprises a first step recessed from the MFS and a second step recessed from the MFS; and a central portion disposed between the first and second side portions at the leading edge, the central portion comprising a notch.
2. The tape head module of claim 1, wherein the notch is recessed from the MFS.
3. The tape head module of claim 1, wherein the notch is disposed at the MFS.
4. A tape drive comprising the tape head module of claim 1.
5. The tape head module of claim 1, wherein the central portion further comprises one or more steps.
6. The tape head module of claim 1, wherein the central portion further comprises a tapered portion.
7. The tape head module of claim 6, wherein the tapered portion is disposed at an angle of about 0.1 degrees to about 2 degrees with respect to the MFS.

8. The tape head module of claim 1, wherein the first step is recessed a distance of about 0.2 μm to about 2.5 μm from the MFS, and wherein the second step is recessed a distance of about 0.5 μm to about 5 μm from the MFS.
 9. A tape head module, comprising: a substrate; a plurality of data heads disposed adjacent to the substrate at a media facing surface (MFS), the plurality of data heads being one or more of write heads and read heads; and a closure disposed adjacent to the plurality of data heads, the closure comprising: a first side portion disposed at a leading edge; a second side portion disposed at the leading edge, wherein the first and second side portions each individually comprises a first step recessed from the MFS and a second step recessed from the MFS; and a central portion disposed between the first and second side portions at the leading edge, the central portion being at least partially recessed from the MFS.
 10. The tape head module of claim 9, wherein the central portion is entirely recessed from the MFS a distance of about 0.1 μm to about 10 μm .
 11. The tape head module of claim 9, wherein the central portion comprises a notch disposed at the MFS and a tapered portion recessed from the MFS.
 12. The tape head module of claim 11, wherein the tapered portion is disposed at an angle of about 0.1 degrees to about 0.5 degrees with respect to the MFS, and wherein the notch has a width of about 1 mm to about 6 mm.
 13. The tape head module of claim 9, wherein the first step is recessed a distance of about 0.2 μm to about 1 μm from the MFS, and wherein the second step is recessed a distance of about 0.5 μm to about 2.5 μm from the MFS.
 14. A tape drive comprising the tape head module of claim 9.
 15. A tape head module, comprising: a substrate; a plurality of data heads disposed adjacent to the substrate at a media facing surface (MFS), the plurality of data heads being one or more of write heads and read heads; and a closure disposed adjacent to the plurality of data heads, the closure comprising: a first side portion disposed at a leading edge; a second side portion disposed at the leading edge, wherein the first and second side portions each individually comprises a first step recessed from the MFS and a second step recessed from the MFS; and a central portion disposed between the first and second side portions at the leading edge, the central portion comprising a third step recessed from the MFS and a fourth step recessed from the MFS.
 16. The tape head module of claim 15, wherein the first and second steps are recessed further from the MFS than the third and fourth steps.
 17. The tape head module of claim 15, wherein the third and fourth steps are recessed further from the MFS than the first and second steps.
 18. The tape head module of claim 15, wherein the first step is recessed a distance of about 0.2 μm to about 1 μm from the MFS, and wherein the second step is recessed a distance of about 0.5 μm to about 2 μm from the MFS.
 19. The tape head module of claim 15, wherein the third step is recessed a distance of about 0.6 μm to about 2.5 μm from the MFS, and wherein the fourth step is recessed a distance of about 1 μm to about 3 μm from the MFS.
 20. The tape head module of claim 15, wherein the third step is recessed a distance of about 0.05 μm to about 0.15 μm from the MFS, and wherein the fourth step is recessed a distance of about 0.155 μm to about 0.2 μm from the MFS.
 21. A tape drive comprising the tape head module of claim 15.
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